MSE – Mean Squared Error

RMSE – Root Mean Squared Error

MAE – Mean Absolute Error

MAPE – Mean Absolute Percentage Error

**Baseline Model:**

A baseline model is a simple model used as a reference to determine whether a more complex model is an improvement.

Example: A model that predicts the mean of the target variable for all observations

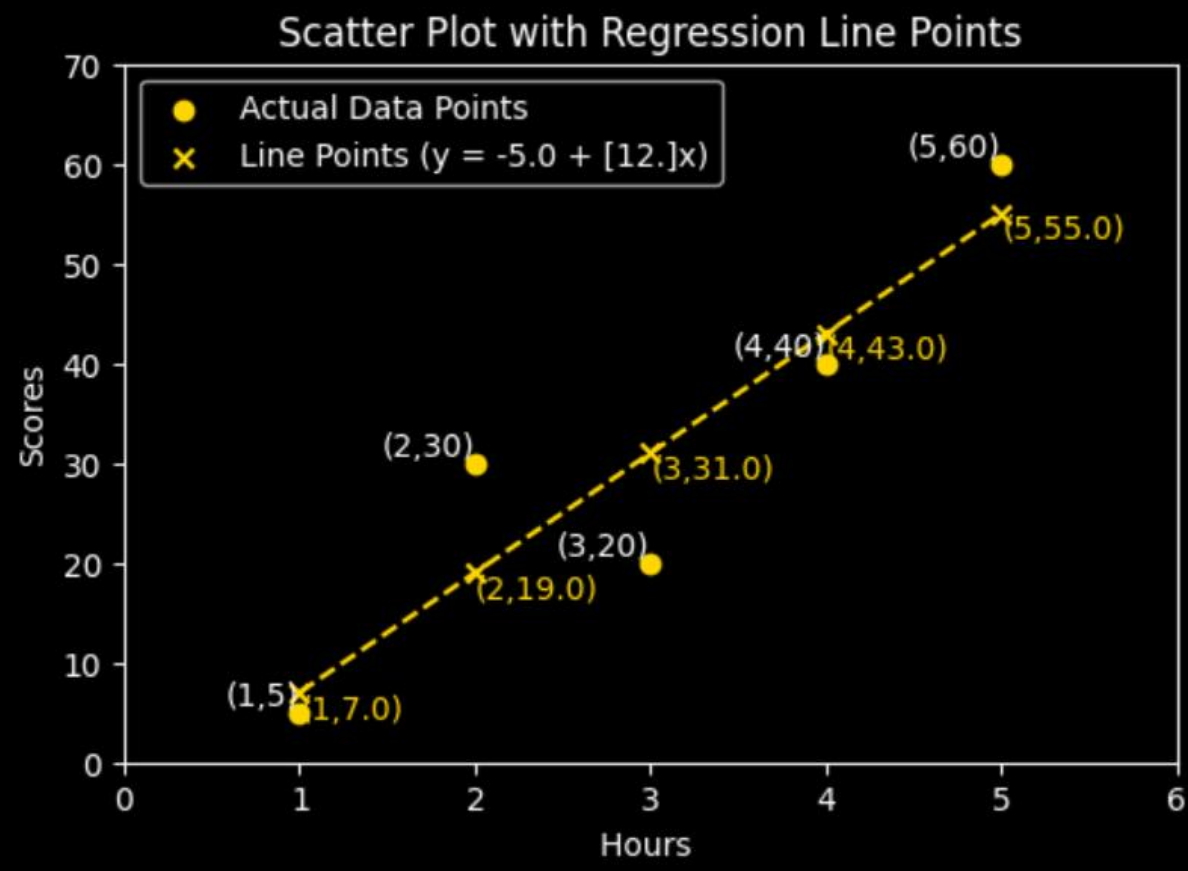
| Hours Studied (x) | Score (y) | Baseline Predicted | Some Model |
|---|---|---|---|
| 1 | 5 | **31** | **7** |
| 2 | 30 | **31** | **37** |
| 3 | 20 | **31** | **24** |
| 4 | 40 | **31** | **35** |
| 5 | 60 | **31** | **53** |
| | Mean = 31 | | |

Any model should give results better than baseline model

# Mean Squared Error:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

| Hours Studied (x) | Score (y) | Model Predict ($\hat{y} = -5 + 12x$) |
|---|---|---|
| 1 | 5 | **7** |
| 2 | 30 | **19** |
| 3 | 20 | **31** |
| 4 | 40 | **43** |
| 5 | 60 | **55** |



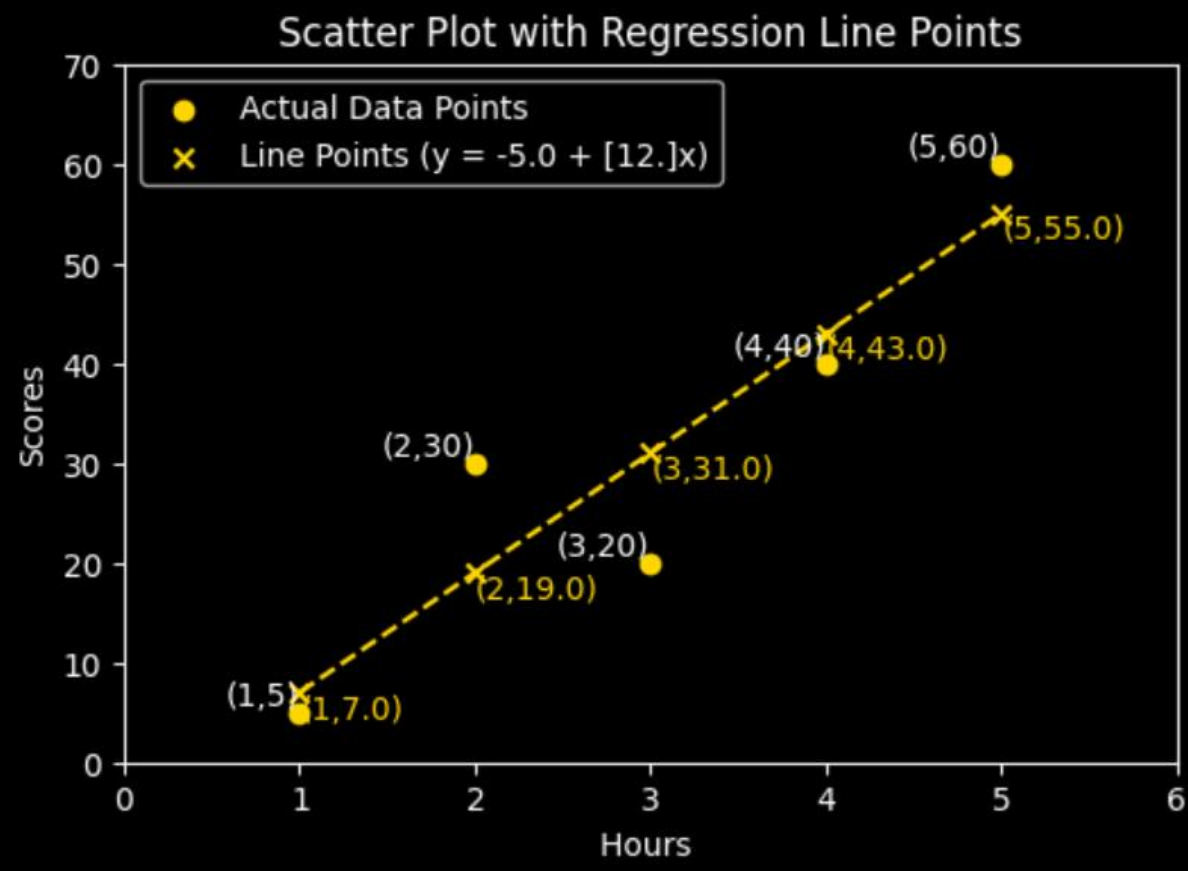

Example:

$$\textbf{MSE} = \frac{[5-7]^2 + [30-19]^2 + \ldots + [60-55]^2\ )}{5} = 56$$

Now let's compare MSE with baseline model:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

| Hours Studied (x) | Score (y) | Model Predict ($\hat{y} = -5 + 12x$) | Baseline Model |
|---|---|---|---|
| 1 | 5 | **7** | **31** |
| 2 | 30 | **19** | **31** |
| 3 | 20 | **31** | **31** |
| 4 | 40 | **43** | **31** |
| 5 | 60 | **55** | **31** |




Example:

**MSE** = $\dfrac{[5-7]^2 + [30-19]^2 + \ldots + [60-55]^2\ )}{5} = 56$

**MSE** Baseline = $\dfrac{[5-31]^2 + [30-31]^2 + \ldots + [60-31]^2\ )}{5} = 344.$

Model is better than baseline

Mean Squared Error:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

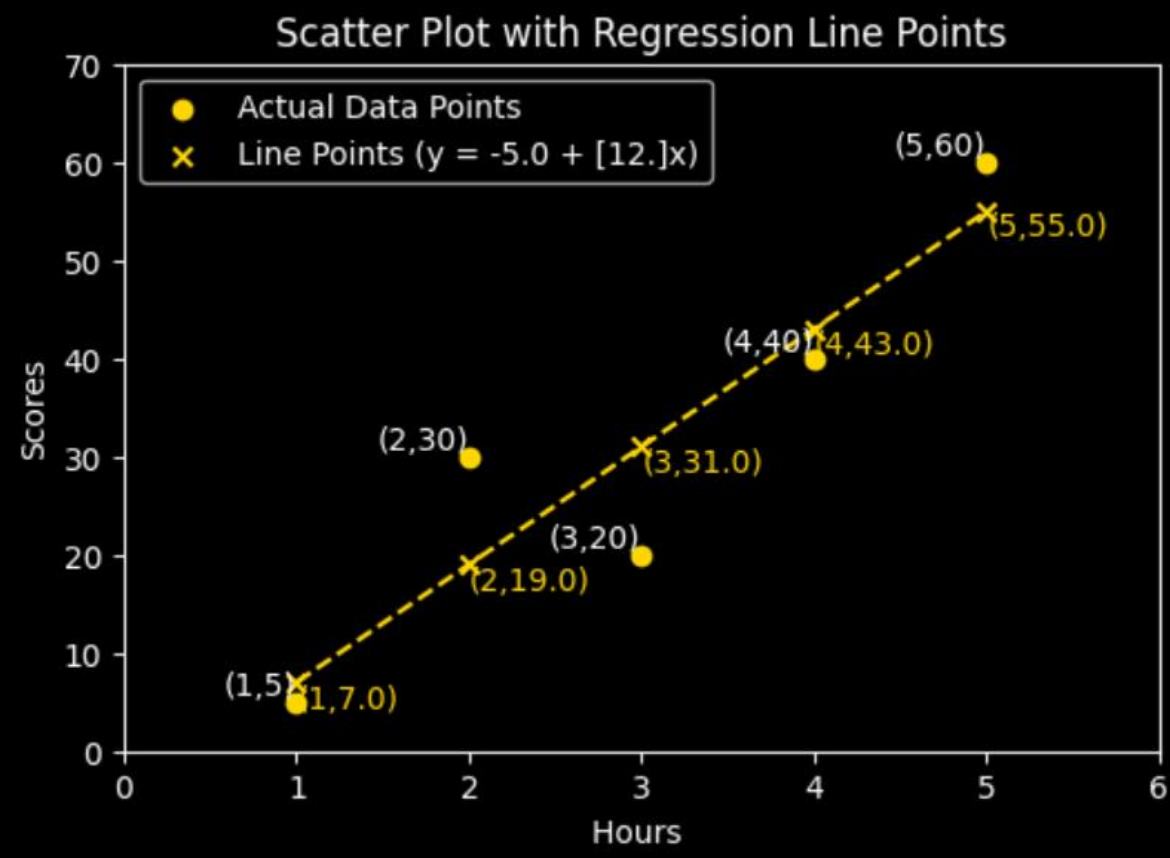
Note:

- The **unit is squared**, so it's harder to interpret directly.
- **Sensitive to large errors,** since errors are squared. Outliers influences its value — a single big error can dominate the loss.
- **Penalizes larger errors more strongly** (because of squaring). If you want your model to care a lot about outliers or big mistakes, MSE is better.
- **Mathematically convenient** — differentiable and smooth everywhere, which makes it easier for gradient-based optimization.

# Root Mean Squared Error:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}$$

| Hours Studied (x) | Score (y) | Model Predict ($\hat{y} = -5 + 12x$) |
|---|---|---|
| 1 | 5 | **7** |
| 2 | 30 | **19** |
| 3 | 20 | **31** |
| 4 | 40 | **43** |
| 5 | 60 | **55** |




Example:

**MSE** = $\dfrac{[5-7]^2 + [30-19]^2 + \ldots + [60-55]^2\ )}{5} = 56$

**RMSE** = $\sqrt{MSE} = \sqrt{56} = 7.48$

# Root Mean Squared Error:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}$$

Note:

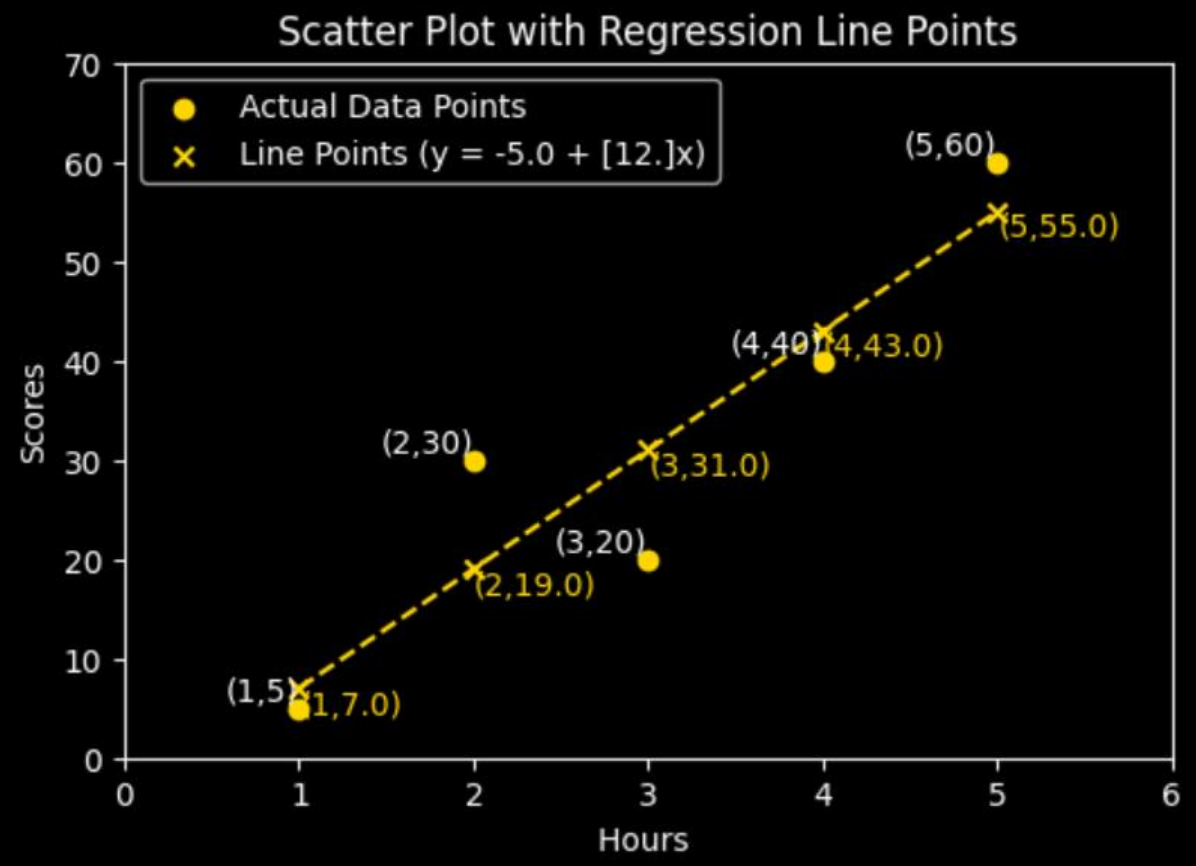
It is in same units as the target variable. So easy to interpret.

**Robust to outliers** — treats all errors linearly, so a few extreme values don't overly affect the model.

Mean Absolute Error:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|Y_i - \hat{Y}_i|$$

| Hours Studied (x) | Score (y) | Model Predict ($\hat{y} = -5 + 12x$) |
|---|---|---|
| 1 | 5 | **7** |
| 2 | 30 | **19** |
| 3 | 20 | **31** |
| 4 | 40 | **43** |
| 5 | 60 | **55** |




Example:

$$\textbf{MAE} = \frac{|5-7| + |30-19| + |20-31| + |40-43| + |60-55|}{5} = 6.2$$

Note: MAE is in the same unit as the target variable.

# Mean Absolute Percentage Error (MAPE) :

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_i - \hat{Y}_i}{Y_i}\right|$$

| Hours Studied (x) | Score (y) | Predicted ($\hat{y} = -5 + 12x$) |
|---|---|---|
| 1 | 5 | **7** |
| 2 | 30 | **19** |
| 3 | 20 | **31** |
| 4 | 40 | **43** |
| 5 | 60 | **55** |

$$MAPE = \frac{1}{5}\left(\left|\frac{5-7}{5}\right| + \left|\frac{30-19}{30}\right| + \cdots + \left|\frac{60-55}{60}\right|\right) = 0.295$$
$$= 29.5\%$$

The model's predictions are off by about **29.5% on average**.

# Mean Absolute Percentage Error (**MAPE**) :

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_i - \hat{Y}_i}{Y_i}\right|$$

2 things to consider when using MAPE:

- Data should not contain zeros (Division by 0 gives error)
- All values should be reasonably large, otherwise it inflate percentage error (division by small number -> big number).

# Mean Absolute Percentage Error (**MAPE**) :

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_i - \hat{Y}_i}{Y_i}\right|$$

| MAPE (%) | Interpretation | Meaning in Plain English |
|---|---|---|
| **< 10%** | Excellent | Predictions are very close to actual values |
| **10% – 20%** | Good | Model is accurate and reliable |
| **20% – 50%** | Reasonable / Moderate | Model has noticeable error but may still be usable |
| **50% – 100% or > 100%** | Poor | Predictions are highly inaccurate |

# EXTRA

## Mean Squared Error (MSE)

Use when:

- You want to **penalize large errors more heavily** (squaring effect).
- The error distribution is expected to be **Gaussian (normal noise)**.
- You want mathematical convenience (e.g., closed-form solution in linear regression).

Best for: **Training models where you want sensitivity to outliers.**

## Mean Absolute Error (MAE)

Use when:

- Data has **outliers or heavy-tailed distributions**, and you don't want them to dominate.
- You want the **typical/median error** rather than being skewed by large mistakes.
- Robustness is more important than punishing big deviations.

Best for: **Noisy or messy datasets** where outliers are common.

## Root Mean Square Error (RMSE)

Use when:

- You want the error in the same units as the target variable (interpretable).
- You care about both sensitivity to large errors (like MSE) and readability.
- You need a metric that’s easier to communicate (e.g., "average prediction is off by 3 units").

Best for: Reporting model performance in real-world units.